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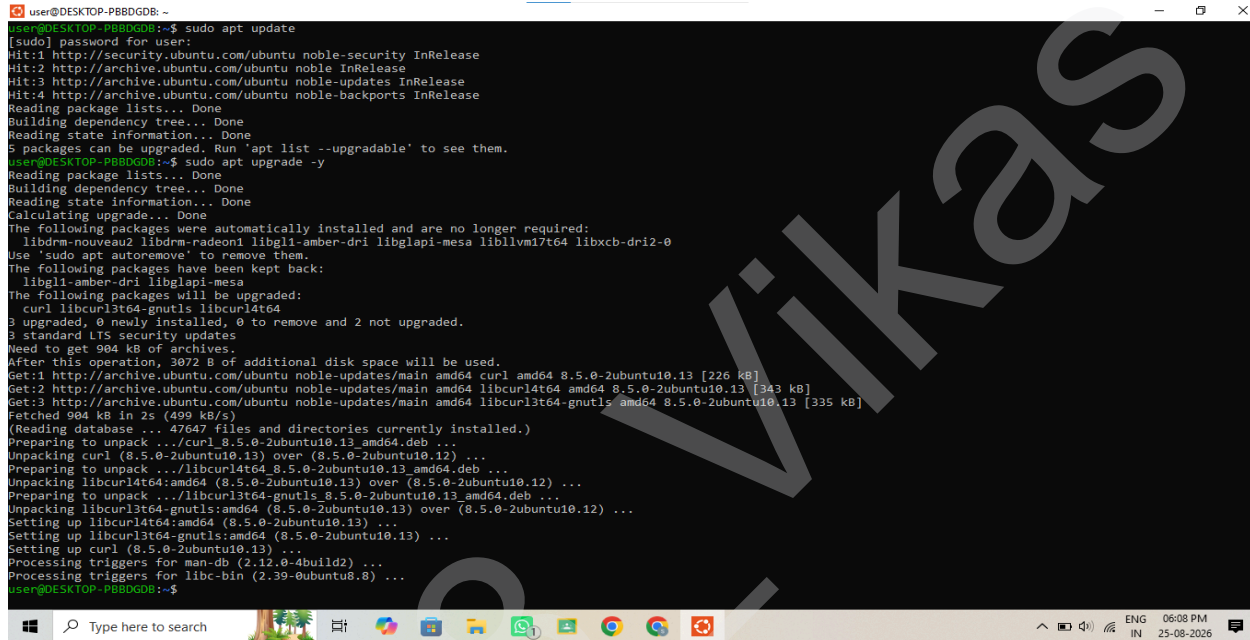
Subject:-Cyber & Information Security

PRACTICAL - 8

Aim: Intrusion Detection System

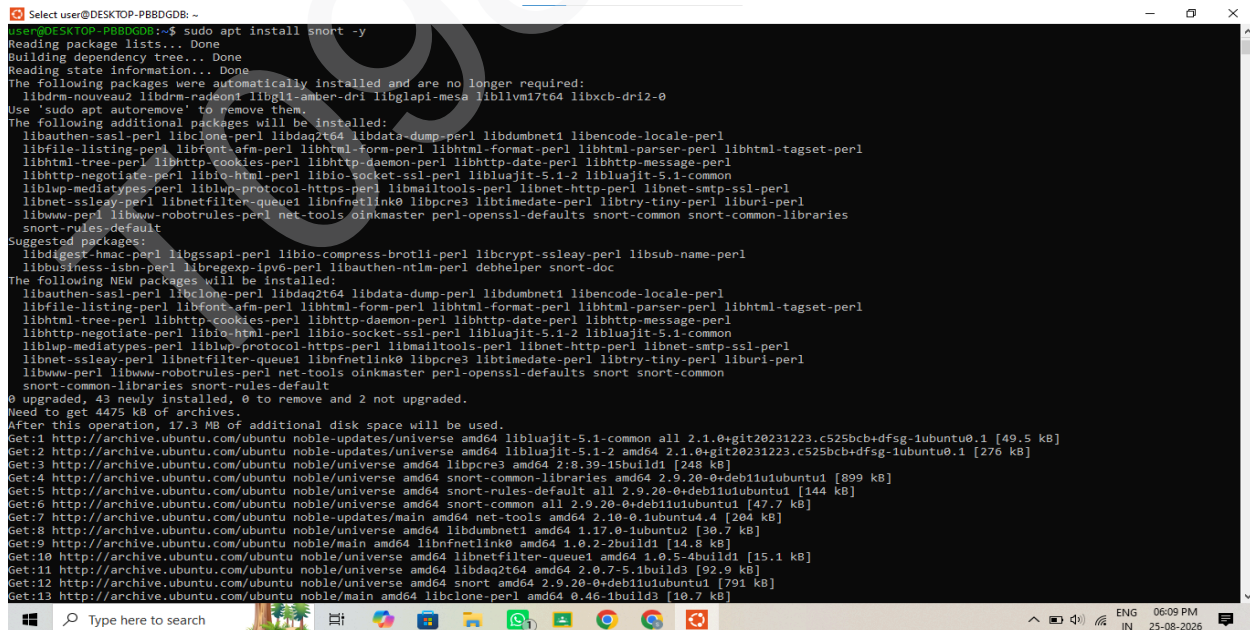
Set up and configure an intrusion detection system (IDS) to monitor network traffic and detect potential security breaches or malicious activities.

Step 1: Open the **Terminal** and run **sudo apt update** followed by **sudo apt upgrade -y** to update Ubuntu and upgrade installed packages.



```
user@DESKTOP-PB8DQDB: ~  
user@DESKTOP-PB8DQDB:~$ sudo apt update  
[sudo] password for user:  
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
5 packages can be upgraded. Run 'apt list --upgradable' to see them.  
user@DESKTOP-PB8DQDB:~$ sudo apt upgrade -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages were automatically installed and are no longer required:  
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-gpu-dri libglapi-mesa libllvm17t64 libxcb-dri2-0  
Use 'sudo apt autoremove' to remove them.  
The following packages have been kept back:  
  libgl1-amd-gpu-dri libglapi-mesa  
The following packages will be upgraded:  
  curl libcurl3t64-gnutls libcurl4t64  
3 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.  
3 standard LTS security updates  
Need to get 904 kB of archives.  
After this operation, 3072 B of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.13 [226 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubuntu10.13 [343 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.13 [335 kB]  
Fetched 904 kB in 2s (499 kB/s)  
(Reading database ... 47647 files and directories currently installed.)  
Preparing to unpack .../curl_8.5.0-2ubuntu10.13_amd64.deb ...  
Unpacking curl (8.5.0-2ubuntu10.13) over (8.5.0-2ubuntu10.12) ...  
Preparing to unpack .../libcurl4t64_8.5.0-2ubuntu10.13_amd64.deb ...  
Unpacking libcurl4t64:amd64 (8.5.0-2ubuntu10.13) over (8.5.0-2ubuntu10.12) ...  
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.13_amd64.deb ...  
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.13) over (8.5.0-2ubuntu10.12) ...  
Setting up libcurl4t64:amd64 (8.5.0-2ubuntu10.13) ...  
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.13) ...  
Setting up curl (8.5.0-2ubuntu10.13) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8) ...  
user@DESKTOP-PB8DQDB:~$
```

Step 2: Open the **Terminal** and run **sudo apt install snort -y** to install Snort.



```
Select user@DESKTOP-PB8DQDB: ~  
user@DESKTOP-PB8DQDB:~$ sudo apt install snort -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-gpu-dri libglapi-mesa libllvm17t64 libxcb-dri2-0  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  libbauthen-sasl-perl libclone-perl libdata-dump-perl libdumbnet1 libencode-locale-perl  
  libfile-listing-perl libfont-afm-perl libhtml-form-perl libhtml-format-perl libhtml-parser-perl libhtml-tagset-perl  
  libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl libhttp-message-perl  
  libhttp-negotiate-perl libio-html-perl libio-socket-ssl-perl liblua5.1-2 liblua5.1-common  
  liblwp-mediatypes-perl liblwp-protocol-http-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl  
  libnet-ssleay-perl libnetfilter-queue1 libnfnetlink0 libnfc3 libnftables-perl libtiny-perl liburi-perl  
  libwww-perl libwww-robotrules-perl net-tools oinkmaster perl-openssl-defaults snort-common snort-common-libraries  
  snort-rules-default  
Suggested packages:  
  libdigest-hmac-perl libgssapi-perl libio-compress-brotli-perl libcrypt-ssleay-perl libsub-name-perl  
  libbusiness-isbn-perl libregexp-ipv6-perl libbauthen-ntlm-perl debhelper snort-doc  
The following NEW packages will be installed:  
  libbauthen-sasl-perl libclone-perl libdata-dump-perl libdumbnet1 libencode-locale-perl  
  libfile-listing-perl libfont-afm-perl libhtml-form-perl libhtml-format-perl libhtml-parser-perl libhtml-tagset-perl  
  libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl libhttp-message-perl  
  libhttp-negotiate-perl libio-html-perl libio-socket-ssl-perl liblua5.1-2 liblua5.1-common  
  liblwp-mediatypes-perl liblwp-protocol-http-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl  
  libnet-ssleay-perl libnetfilter-queue1 libnfnetlink0 libnfc3 libnftables-perl libtiny-perl liburi-perl  
  libwww-perl libwww-robotrules-perl net-tools oinkmaster perl-openssl-defaults snort snort-common  
  snort-common-libraries snort-rules-default  
0 upgraded, 43 newly installed, 0 to remove and 2 not upgraded.  
Need to get 4475 kB of archives.  
After this operation, 17.3 MB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 liblua5.1-2 amd64 2.1.0+git20231223.c525bcb+dfsg-1ubuntu0.1 [49.5 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 liblua5.1-2 amd64 2.1.0+git20231223.c525bcb+dfsg-1ubuntu0.1 [276 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnfc3 amd64 2.8.39-1build1 [248 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-common-libraries amd64 2.9.20-0+deb11u1ubuntu1 [899 kB]  
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-rules-default all 2.9.20-0+deb11u1ubuntu1 [144 kB]  
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnetfilter-queue1 amd64 2.9.20-0+deb11u1ubuntu1 [47.7 kB]  
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 net-tools amd64 2.10.0-1ubuntu4.4 [204 kB]  
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 libdumbnet1 amd64 1.17.0-1ubuntu2 [30.7 kB]  
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 libnfnetlink0 amd64 1.0.2-2build1 [14.8 kB]  
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnetfilter-queue1 amd64 1.0.5-4build1 [15.1 kB]  
Get:11 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnftables-perl amd64 2.0.7-5.1build3 [92.9 kB]  
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort amd64 2.9.20-0+deb11u1ubuntu1 [791 kB]  
Get:13 http://archive.ubuntu.com/ubuntu noble/main amd64 libclone-perl amd64 0.46-1build3 [10.7 kB]
```

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Step 3: Run `ip addr` in the **Terminal** to view and identify the available network interfaces and their IP addresses.

```
user@DESKTOP-PB8DGD8: ~  
Processing triggers for man-db (2.12.0-4build2) ...  
user@DESKTOP-PB8DGD8:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: bond0: <BROADCAST,MULTICAST,MASTER> mtu 1500 qdisc noop state DOWN group default qlen 1000  
    link/ether 4e:b7:34:fc:c8:4e brd ff:ff:ff:ff:ff:ff  
3: dummy0: <BROADCAST,NOARP> mtu 1500 qdisc noop state DOWN group default qlen 1000  
    link/ether 62:75:57:6d:8d:d5 brd ff:ff:ff:ff:ff:ff  
4: tunl0@NONE: <NOARP> mtu 1480 qdisc noop state DOWN group default qlen 1000  
    link/ipip 0.0.0.0 brd 0.0.0.0  
5: sit0@NONE: <NOARP> mtu 1480 qdisc noop state DOWN group default qlen 1000  
    link/sit 0.0.0.0 brd 0.0.0.0  
6: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000  
    link/ether 00:15:5d:29:16:0b brd ff:ff:ff:ff:ff:ff  
    inet 172.29.188.110/20 brd 172.29.191.255 scope global eth0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::215:5dff:fe29:160b/64 scope link  
        valid_lft forever preferred_lft forever  
user@DESKTOP-PB8DGD8:~$
```

Step 4: Run `sudo snort -T -c /etc/snort/snort.conf` in the **Terminal** to test and verify the Snort configuration.

```
user@DESKTOP-PB8DGD8: ~  
Verifying Preprocessor Configurations!  
WARNING: flowbits key 'smb.tree.create.llsrpc' is set but not ever checked.  
WARNING: flowbits key 'ms_sql_dns' is checked but not ever set.  
33 out of 1024 flowbits in use.  
MaxRss at the end of rules:65084  
[ Port Based Pattern Matching Memory ]  
[ Aho-Corasick Summary ] -----  
Storage Format : Full-Q  
Finite Automaton : DFA  
Alphabet Size : 256 Chars  
Sizeof State : Variable (1,2,4 bytes)  
Instances : 215  
1 byte states : 204  
2 byte states : 11  
4 byte states : 0  
Characters : 64755  
States : 31951  
Transitions : 863868  
State Density : 10.6%  
Patterns : 5041  
Match States : 3836  
Memory (MB) : 16.90  
Patterns : 0.51  
Match Lists : 1.01  
DFA  
1 byte states : 1.02  
2 byte states : 13.96  
4 byte states : 0.00  
-----  
[ Number of patterns truncated to 20 bytes: 1038 ]  
MaxRss at the end of detection rules:108588  
---- Initialization Complete ----  
--> Snort! <--  
Version 2.0.20 GRE (Build 82)  
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team  
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.  
Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
Using libpcap version 1.10.4 (with TPACKET_V3)  
Using PCRE version: 8.39 2016-06-14
```

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```
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Total snort Fixed Memory Cost - MaxRss:108588
Snort successfully validated the configuration!
Snort exiting
user@DESKTOP-P8BDGDB:~$
```

Step 5: Run `sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf`, replacing **enp0s3** with your actual interface, to start Snort in network monitoring mode and display alerts in the terminal.

```
user@DESKTOP-P8BDGDB: ~
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
ERROR: /etc/snort/snort.conf(0) Unable to open rules file "/etc/snort/snort.conf": No such file or directory.

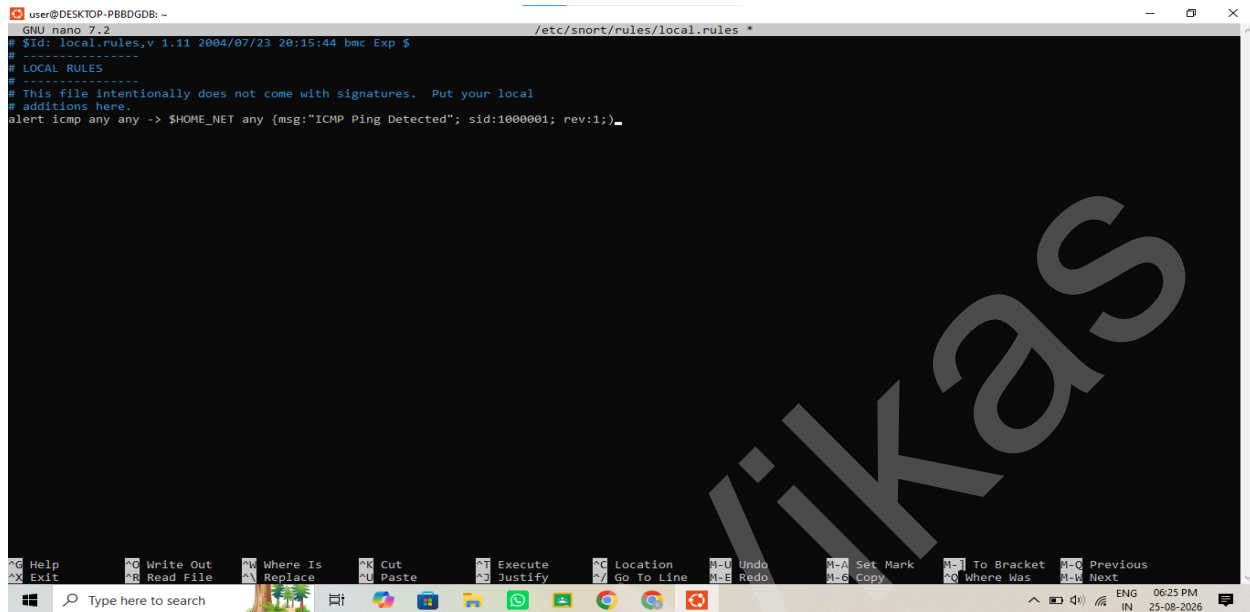
Fatal Error, Quitting..
user@DESKTOP-P8BDGDB:~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Running in IDS mode

---- Initializing Snort ----
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 501 593 901 1220 1414 1741 1830 2381 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 800
9 8080 8014 8020 8080 8085 8088 8090 8110 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50802 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5081 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 501 593 901 1220 1414 1741 1830 2381 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510
7777 7779 8000 8080 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50802 555
55 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-0
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
  Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsf_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
Finished Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dce2_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_s7complus_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_reputation_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ftotelnet_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_gtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssl_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_modbus_preproc.so... done
```

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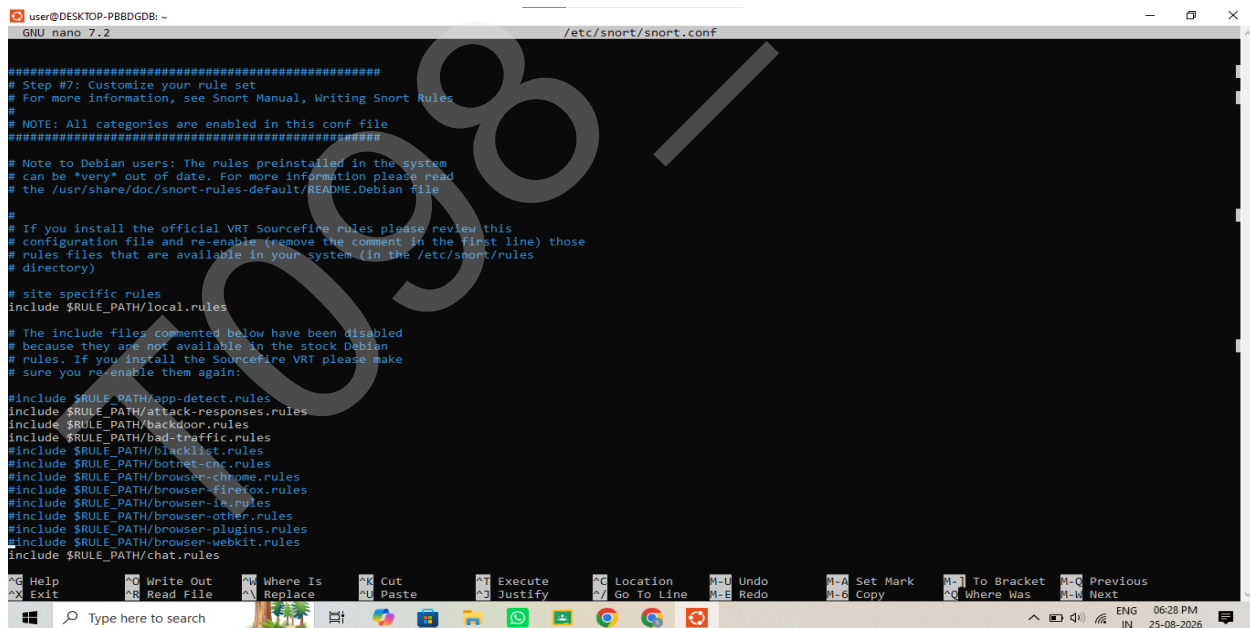
Subject:-Cyber & Information Security

Step 6: Open `/etc/snort/rules/local.rules` with `sudo nano`, add the **ICMP Ping Detected** alert rule with **SID 1000001**, then save the file to detect ICMP ping traffic.



```
GNU nano 7.2 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#-----
# LOCAL RULES
#-----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:'ICMP Ping Detected'; sid:1000001; rev:1);
```

Step 7: Open `/etc/snort/snort.conf` and verify that the line `include $RULE_PATH/local.rules` is present to ensure the local rules file is loaded.



```
GNU nano 7.2 /etc/snort/snort.conf
# $Id: snort.conf,v 1.11 2004/07/23 20:15:44 bmc Exp $
#-----
# Step #7: Customize your rule set
# For more information, see Snort Manual, Writing Snort Rules
# NOTE: All categories are enabled in this conf file
#-----
# Note to Debian users: The rules preinstalled in the system
# can be *very* out of date. For more information please read
# the /usr/share/doc/snort-rules-default/README.Debian file
#
# If you install the official VRT Sourcefire rules please review this
# configuration file and re-enable (remove the comment in the first line) those
# rules files that are available in your system (in the /etc/snort/rules
# directory)
#
# site specific rules
include $RULE_PATH/local.rules
#
# The include files commented below have been disabled
# because they are not available in the stock Debian
# rules. If you install the Sourcefire VRT please make
# sure you re-enable them again:
#
# include $RULE_PATH/app-detect.rules
# include $RULE_PATH/attack-responses.rules
# include $RULE_PATH/backdoor.rules
# include $RULE_PATH/bad-traffic.rules
# include $RULE_PATH/ba-wklist.rules
# include $RULE_PATH/bother-cnc.rules
# include $RULE_PATH/browser-chrome.rules
# include $RULE_PATH/browser-firefox.rules
# include $RULE_PATH/browser-ie.rules
# include $RULE_PATH/browser-other.rules
# include $RULE_PATH/browser-plugins.rules
# include $RULE_PATH/browser-webkit.rules
# include $RULE_PATH/chat.rules
```

Step 8: Run `sudo snort -T -c /etc/snort/snort.conf` again and fix any configuration errors before proceeding.

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Subject:-Cyber & Information Security

```
user@DESKTOP-PB8DGD8: ~
Patterns      : 0.51
Match Lists   : 1.01
DFA
1 byte states : 1.02
2 byte states : 13.96
4 byte states : 0.00
-----
[ Number of patterns truncated to 20 bytes: 1038 ]
MaxRss at the end of detection rules:106888
---- Initialization Complete ----

-*> Snort! <-*
o''~
...~
Version 2.9.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_INAP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTTP Version 1.1 <Build 9>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_WOBBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCEPC2 Version 1.0 <Build 3>

Total snort Fixed Memory Cost - MaxRss:106888
Snort successfully validated the configuration!
Snort exiting
user@DESKTOP-PB8DGD8:~$ ip addr
```

Step 9: Run `sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf` (replace **enp0s3** with your interface) to start active IDS monitoring and display alerts in the terminal.

```
user@DESKTOP-PB8DGD8: ~
Patterns      : 0.51
Match Lists   : 1.01
DFA
1 byte states : 1.02
2 byte states : 13.96
4 byte states : 0.00
-----
[ Number of patterns truncated to 20 bytes: 1038 ]
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x7f98c0b026c0 (1208)
Decoding Ethernet
---- Initialization Complete ----

-*> Snort! <-*
o''~
...~
Version 2.9.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_INAP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTTP Version 1.1 <Build 9>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_WOBBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCEPC2 Version 1.0 <Build 3>

commencing packet processing (pid=1207)
```

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Subject:-Cyber & Information Security

```
user@DESKTOP-PB8DGB: ~
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_PIPELINET Version 1.2 <Build 13>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_S7COMMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCEPC2 Version 1.0 <Build 3>
Commencing packet processing (pid=1207)
Exit
08/25-13:30:53.954312 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:30:55.085781 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:30:56.045798 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:30:57.085744 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:30:58.125725 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:30:59.165636 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:00.205690 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:01.245701 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:02.285703 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:03.325699 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:04.366142 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:05.405400 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:06.445481 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:07.485707 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:08.525446 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:09.565488 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:10.605470 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:11.645699 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:12.685528 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:13.725462 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:14.765621 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:15.805581 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:16.845512 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:17.885505 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:18.925508 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:19.965503 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:21.005451 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:22.045533 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:23.085492 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:24.125485 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:25.165482 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:26.205455 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:27.245715 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
08/25-13:31:28.285476 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.29.188.110 -> 172.19.188.10
```

Step 10: Open the Snort configuration file by running `sudo nano /etc/snort/snort.conf` in the Terminal.

Step 11: Run `sudo apt install nmap -y` in the Terminal to install Nmap for testing port-scan detection.

```
user@DESKTOP-PB8DGB: ~
Install: Invalid option -- '-y'
Try 'install --help' for more information.
user@DESKTOP-PB8DGB:~$ sudo apt install nmap -y
[sudo] password for user:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgll-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libblas3 liblinear4 libssh2-t64 nmap-common
Suggested packages:
  liblinear-tools liblinear-dev ncat ndiff zenmap
The following NEW packages will be installed:
  libblas3 liblinear4 libssh2-t64 nmap nmap-common
0 upgraded, 5 newly installed, 0 to remove and 2 not upgraded.
Need to get 6287 kB of archives.
After this operation, 27.4 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libblas3 amd64 3.12.0-3build1.1 [238 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 liblinear4 amd64 2.3.0+dfsg-5build1 [42.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libssh2-t64 amd64 1.11.0-4.1ubuntu0.24.04.3 [121 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap-common all 7.94+git20230807.3be01efb1+dfsg-3build2 [4192 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap amd64 7.94+git20230807.3be01efb1+dfsg-3build2 [1694 kB]
Fetched 6287 kB in 4s (1500 kB/s)
Selecting previously unselected package libblas3:amd64.
(Reading database ... 48806 files and directories currently installed.)
Preparing to unpack .../libblas3:amd64 (3.12.0-3build1.1) ...
Unpacking libblas3:amd64 (3.12.0-3build1.1) ...
Selecting previously unselected package liblinear4:amd64.
Preparing to unpack .../liblinear4:amd64 (2.3.0+dfsg-5build1) ...
Unpacking liblinear4:amd64 (2.3.0+dfsg-5build1) ...
Selecting previously unselected package libssh2-t64:amd64.
Preparing to unpack .../libssh2-t64:amd64 (1.11.0-4.1ubuntu0.24.04.3) ...
Unpacking libssh2-t64:amd64 (1.11.0-4.1ubuntu0.24.04.3) ...
Selecting previously unselected package nmap-common.
Preparing to unpack .../nmap-common:amd64 (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Unpacking nmap-common (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Selecting previously unselected package nmap.
Preparing to unpack .../nmap:amd64 (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Unpacking nmap (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Setting up libblas3:amd64 (3.12.0-3build1.1) ...
update-alternatives: using /usr/lib/x86_64-linux-gnu/libblas.so.3 to provide /usr/lib/x86_64-linux-gnu/libblas.so.3 (libblas.so.3-x86_64-linux-gnu) in auto mode
Setting up nmap-common (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Setting up libssh2-t64:amd64 (1.11.0-4.1ubuntu0.24.04.3) ...
Setting up nmap (7.94+git20230807.3be01efb1+dfsg-3build2) ...
```

Add the **TCP SYN scan detection** rule with **SID 1000002** to `/etc/snort/rules/local.rules`, restart Snort, then run `nmap -sS <TARGET-IP-ADDRESS>` from an authorized lab machine to test detection.

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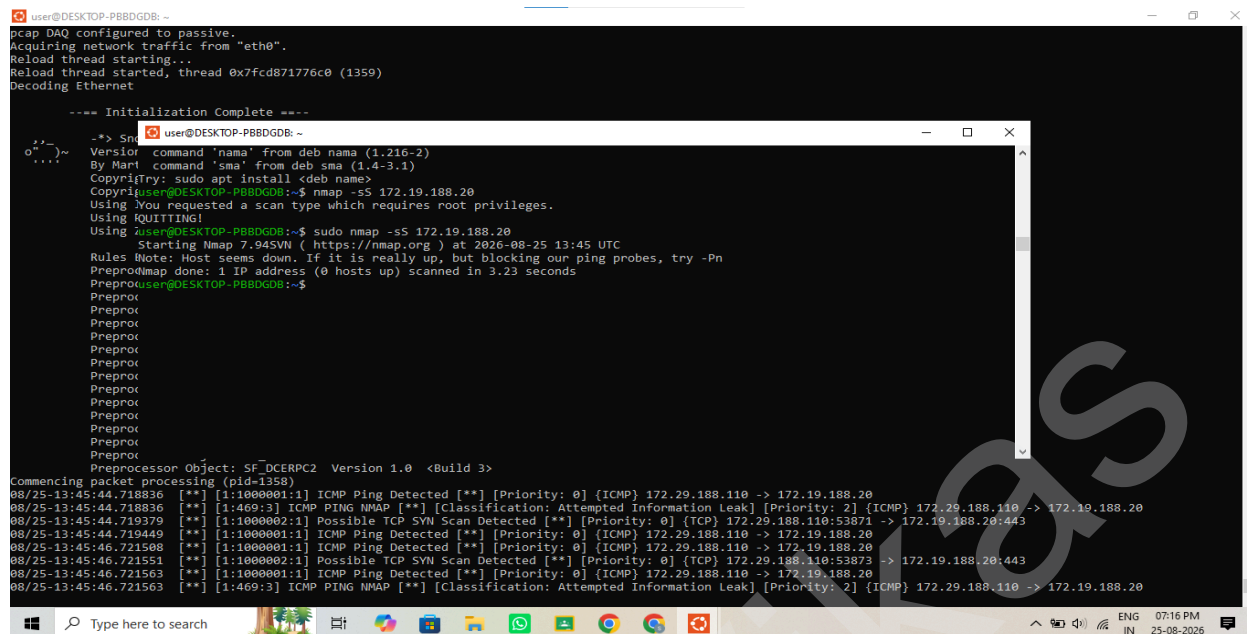
Subject:-Cyber & Information Security

```
user@DESKTOP-PBBDGDB: ~  
GNU nano 7.2 /etc/snort/rules/local.rules *  
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $  
# -----  
# LOCAL RULES  
# -----  
# This file intentionally does not come with signatures. Put your local  
# additions here.  
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)  
alert tcp any any -> $HOME_NET any (flags:S; msg:"Possible TCP SYN Scan Detected"; sid:1000002; rev:1;)  
  
File Name to Write: /etc/snort/rules/local.rules  
^C Help      ^O DOS Format  ^N-A Append  ^N-B Backup File  
^G Cancel    ^M Mac Format  ^N-P Prepend  ^T Browse
```

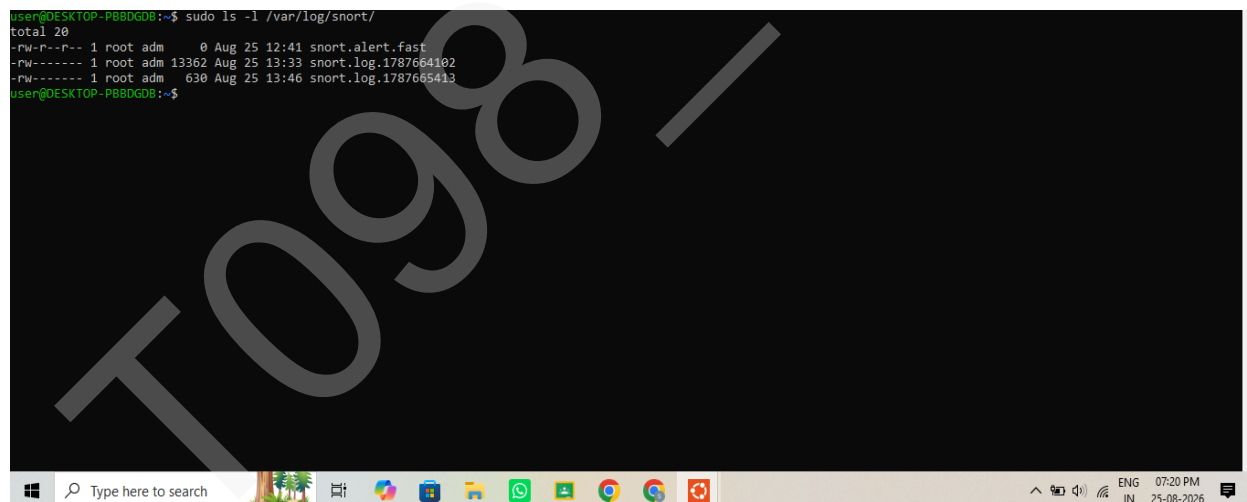
```
user@DESKTOP-PBBDGDB: ~  
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>  
Preprocessor Object: SF_DNS Version 1.1 <Build 4>  
Preprocessor Object: SF_SDF Version 1.1 <Build 1>  
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>  
Preprocessor Object: SF_POP Version 1.0 <Build 1>  
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>  
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>  
Preprocessor Object: appid Version 1.1 <Build 5>  
Preprocessor Object: SF_SSH Version 1.1 <Build 3>  
Preprocessor Object: SF_SIP Version 1.1 <Build 1>  
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>  
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>  
Preprocessor Object: SF_GTP Version 1.1 <Build 1>  
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>  
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>  
Preprocessor Object: SF_5TCONWPLUS Version 1.0 <Build 1>  
Preprocessor Object: SF_DCEPRPC2 Version 1.0 <Build 3>  
  
Total snort Fixed Memory Cost - MaxRss:108792  
Snort successfully validated the configuration!  
Snort exiting  
user@DESKTOP-PBBDGDB:~$
```

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Step 12: Check Snort alerts in the terminal using `sudo ls /var/log/snort/` and `sudo cat /var/log/snort/alert`, or view them directly when running Snort with `-A console`.



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Subject:-Cyber & Information Security

Step 13: Run `sudo apt install wireshark -y` in the Terminal to install **Wireshark** for network traffic monitoring and packet analysis.

```
user@DESKTOP-P8BDGDB: ~
Get:73 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-icccm4 amd64 0.4.1-1.1build3 [10.8 kB]
Get:74 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-util1 amd64 0.4.0-1build3 [10.7 kB]
Get:75 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-image0 amd64 0.4.0-2build1 [10.8 kB]
Get:76 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-keysyms1 amd64 0.4.0-1build4 [7956 B]
Get:77 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-render-util0 amd64 0.3.9-1build4 [9608 B]
Get:78 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-shape0 amd64 1.15-1ubuntu2 [6100 B]
Get:79 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-xkb1 amd64 1.15-1ubuntu2 [32.3 kB]
Get:80 http://archive.ubuntu.com/ubuntu noble/main amd64 libxkbcommon-x11-0 amd64 1.6.0-1build1 [14.5 kB]
Get:81 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6gui6 amd64 6.4.2+dfsg-21.1build5 [3119 kB]
Get:82 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6network6 amd64 6.4.2+dfsg-21.1build5 [740 kB]
Get:83 http://archive.ubuntu.com/ubuntu noble/universe amd64 libswscale7 amd64 7:6.1.1-3ubuntu5 [193 kB]
Get:84 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6multimedia6 amd64 6.4.2-11build3 [527 kB]
Get:85 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6opengl6 amd64 6.4.2+dfsg-21.1build5 [420 kB]
Get:86 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6widgets6 amd64 6.4.2+dfsg-21.1build5 [2648 kB]
Get:87 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6printsupport6 amd64 6.4.2+dfsg-21.1build5 [222 kB]
Get:88 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6qml6 amd64 6.4.2+dfsg-4build3 [1619 kB]
Get:89 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6qmlmodels6 amd64 6.4.2+dfsg-4build3 [267 kB]
Get:90 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6quick6 amd64 6.4.2+dfsg-4build3 [2001 kB]
Get:91 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6svg6 amd64 6.4.2-4ubuntu3 [156 kB]
Get:92 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandclient6 amd64 6.4.2-5build3 [251 kB]
Get:93 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandcompositore6 amd64 6.4.2-5build3 [427 kB]
Get:94 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandeglclientintegration6 amd64 6.4.2-5build3 [19.5 kB]
Get:95 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandeglcompositorhwintegration6 amd64 6.4.2-5build3 [15.2 kB]
Get:96 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6wlshellintegration6 amd64 6.4.2-5build3 [38.0 kB]
Get:97 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbsml2t64 amd64 0.4.8+dfsg2-16.1build1 [112 kB]
Get:98 http://archive.ubuntu.com/ubuntu noble/universe amd64 libspandsp2t64 amd64 0.6.6+dfsg-2.1build1 [311 kB]
Get:99 http://archive.ubuntu.com/ubuntu noble/main amd64 libspeexdsp1 amd64 1.2.1-1ubuntu3 [43.1 kB]
Get:100 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwireshark-data all 4.2.2-1.1build3 [454 kB]
Get:101 http://archive.ubuntu.com/ubuntu noble/main amd64 libbsc1 amd64 2.0-1build1 [33.9 kB]
Get:102 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwsutil15t64 amd64 4.2.2-1.1build3 [121 kB]
Get:103 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwireshark1t64 amd64 4.2.2-1.1build3 [259 kB]
Get:104 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwireshark1t64 amd64 4.2.2-1.1build3 [22.7 kB]
Get:105 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 mesa-va-drivers amd64 25.2.8-0ubuntu0.24.04.2 [6784 B]
Get:106 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 mesa-vdpau-drivers amd64 25.2.8-0ubuntu0.24.04.2 [23.2 kB]
Get:107 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-gtk-platformtheme amd64 6.4.2+dfsg-21.1build5 [95.1 kB]
Get:108 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-qpas-plugins amd64 6.4.2+dfsg-21.1build5 [80.3 kB]
Get:109 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-translations-l10n all 6.4.2-1 [1761 kB]
Get:110 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-wayland amd64 6.4.2-5build3 [127 kB]
Get:111 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 1965-va-driver amd64 2.4.1+dfsg1-1ubuntu0.2 [332 kB]
Get:112 http://archive.ubuntu.com/ubuntu noble/universe amd64 va-driver-all amd64 2.20.0-2ubuntu0.2 [5014 B]
Get:113 http://archive.ubuntu.com/ubuntu noble/main amd64 vdpau-driver-all amd64 1.5-2build1 [4414 B]
Get:114 http://archive.ubuntu.com/ubuntu noble/universe amd64 wireshark-common amd64 4.2.2-1.1build3 [587 kB]
95% [114 wireshark-common 20.0 kB/587 kB 3%]
2235 kB/s 2s
```

```
user@DESKTOP-P8BDGDB: ~
Package configuration

Configuring wireshark-common

Dumpcap can be installed in a way that allows members of the "wireshark" system group to capture packets. This is recommended over the alternative of running
Wireshark/Tshark directly as root, because less of the code will run with elevated privileges.

For more detailed information please see /usr/share/doc/wireshark-common/README.Debian.gz once the package is installed.

Enabling this feature may be a security risk, so it is disabled by default. If in doubt, it is suggested to leave it disabled.

Should non-superusers be able to capture packets?
<Yes> <No>
```

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Subject:-Cyber & Information Security

Capturing from vEthernet (WSL)

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	icmp	Source	Destination	Protocol	Length	Info
6	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 6)
7	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 5)
8	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 8)
9	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 7)
10	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) request id=0x0001, seq=35/8960, ttl=128 (reply in 10)
11	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) reply id=0x0001, seq=35/8960, ttl=64 (request in 9)
12	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) request id=0x0001, seq=36/9216, ttl=128 (reply in 12)
13	icmpv6	172.29.176.1	172.29.188.110	ICMP	74	Echo (ping) reply id=0x0001, seq=36/9216, ttl=64 (request in 11)

> Frame 5: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{...} Ethernet II, Src: Microsoft_d1:34:9f (00:15:5d:d1:34:9f), Dst: Microsoft_29:16:00 (00:15:5d:29:16:00) Internet Protocol Version 4, Src: 172.29.176.1, Dst: 172.29.188.110 Internet Control Message Protocol

0000 00 15 5d 29 16 00 00 15 5d d1 34 9f 00 00 45 00 ...]...]... E-
0010 00 3c 92 1d 00 00 80 01 00 00 ac 1d b0 01 ac 1d ... <... ..
0020 bc 6e 08 00 4d 3a 00 01 00 21 61 62 63 64 65 66 ... n: M: ... labcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ... ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 ... wabcedfg hi

Internet Control Message Protocol: Protocol

Packets: 16 - Displayed: 8 (50.0%)

Profile: Default

08:51 PM 25-08-2026